

SEEHAUSEN ALTMARK, Germany

WMO: 102610

Lat: 52.8911N

Lon: 11.7297E

Elev: 76

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.6	13.0	3.4	6.6	9.6	8.6	8.6	13.7	27.6	42.4	24.2	41.6	2.7	110	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	85.3	65.9	81.7	64.5	78.2	63.2	68.3	79.3	66.4	76.9	64.8	74.9	7.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.9	71.6	62.7	85.5	70.0	60.9	80.0	68.8	32.6	78.8	31.1	77.4	29.8	74.4	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	19.9	17.6	DB	5.2	91.5	11.3	3.5	-2.9	94.0	-9.5	96.1	-15.9	98.1	-24.2	100.6	(4)
(5)			WB	4.8	72.5	11.2	2.0	-3.3	73.9	-9.8	75.1	-16.1	76.2	-24.2	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	32.9	32.3	39.6	45.6	54.8	60.0	64.1	63.5	57.0	49.1	39.9	35.7		
	DBStd	13.29	10.60	9.34	6.77	6.26	6.39	6.07	5.47	5.33	4.83	6.65	6.54	7.26		
	HDD50	2389	532	497	324	162	26	1	0	0	3	95	305	444		
	HDD65	6353	996	916	787	583	321	173	85	93	246	493	752	908		
	CDD50	1651	1	1	3	29	174	303	436	420	212	68	3	1		
	CDD65	140	0	0	0	0	4	25	57	48	5	1	0	0		
	CDH74	1829	0	0	0	3	120	372	720	545	64	5	0	0		
	CDH80	485	0	0	0	0	9	99	211	160	6	0	0	0		
	WSAvg	7.5	9.1	8.0	8.3	7.3	6.9	6.9	6.8	6.3	7.0	7.3	8.0	8.6		
	PrecAvg	23.0	1.9	1.5	1.5	1.2	1.9	2.2	2.8	2.4	2.0	1.8	1.8	1.9		
(15) Precipitation	PrecMax	30.9	4.4	3.0	3.8	3.0	5.1	3.9	5.6	5.7	5.4	4.8	3.9	3.2		
	PrecMin	16.0	0.1	0.4	0.3	0.1	0.5	0.8	0.6	0.6	0.4	0.5	0.1	0.7		
	PrecStd	3.7	0.9	0.6	0.9	0.7	1.1	0.9	1.3	1.2	1.1	1.0	0.9	0.7		
	PrecStd	3.7	0.9	0.6	0.9	0.7	1.1	0.9	1.3	1.2	1.1	1.0	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	58.0	63.2	72.4	81.0	88.9	89.0	89.3	80.5	71.9	56.4	54.3		
		MCWB	49.0	51.9	51.6	56.7	61.5	65.7	68.2	69.6	63.6	61.8	51.4	50.3		
	2%	DB	49.4	51.9	57.0	66.5	77.5	82.6	85.5	84.4	75.1	66.4	53.7	51.6		
		MCWB	46.4	47.5	49.6	53.8	59.7	63.7	66.1	66.1	62.7	58.5	50.4	48.8		
	5%	DB	47.2	47.1	52.5	62.1	73.8	78.2	82.0	80.3	70.4	62.1	51.7	49.1		
		MCWB	44.6	43.9	46.6	51.4	58.5	61.6	65.2	64.9	60.1	55.5	49.1	47.0		
	10%	DB	44.9	43.7	49.6	57.6	69.7	74.0	77.9	75.8	66.7	59.0	49.3	46.7		
		MCWB	42.4	40.8	45.3	49.2	56.8	60.7	64.1	63.2	58.6	54.0	47.0	45.0		
	0.4%	WB	49.8	52.3	53.4	58.1	64.4	69.5	71.2	70.9	66.3	63.5	52.9	51.2		
		MCDB	52.3	57.0	61.4	69.9	75.4	76.2	83.9	85.2	76.4	70.3	55.2	53.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.3	47.6	50.4	54.9	61.9	66.0	68.8	67.9	64.1	59.6	51.1	49.2		
		MCDB	49.2	51.3	55.8	63.7	74.0	77.5	80.7	78.1	72.3	65.0	53.3	51.2		
	5%	WB	44.9	44.2	48.0	52.5	59.8	63.8	66.6	66.3	62.0	56.7	49.2	47.0		
		MCDB	46.8	46.9	51.4	59.9	71.1	75.1	78.2	75.6	68.6	60.9	51.4	48.7		
	10%	WB	42.3	41.0	45.5	50.4	57.7	61.6	64.6	64.6	59.7	54.6	47.2	44.8		
		MCDB	44.6	43.3	49.2	57.1	67.9	71.7	75.3	73.8	65.4	58.1	49.1	46.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	11.3	13.2	17.7	20.4	18.8	20.0	19.4	16.7	14.2	9.6	7.8		
		MCDBR	9.7	14.0	19.3	26.0	28.6	28.4	29.0	27.9	24.5	19.4	11.8	9.9		
		MCWBR	8.3	10.8	13.0	14.6	14.2	12.6	13.0	13.0	13.7	12.4	9.7	8.9		
		MCDBR	9.4	13.8	17.9	22.7	24.9	23.7	25.2	22.3	21.0	17.3	11.3	9.4		
	5% WB	MCWBR	8.5	11.1	12.9	13.7	13.5	11.7	13.0	11.5	12.7	12.0	9.7	8.9		
		MCWBR	8.5	11.1	12.9	13.7	13.5	11.7	13.0	11.5	12.7	12.0	9.7	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.340	0.376	0.413	0.411	0.408	0.428	0.420	0.386	0.362	0.338	0.311		
	taud		2.339	2.326	2.251	2.201	2.254	2.282	2.241	2.274	2.353	2.396	2.382	2.337		
	Ebn at Noon		209	240	254	259	266	268	259	254	249	230	200	188		
	Edh at Noon		19	26	34	41	41	40	41	37	31	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		197	417	768	1267	1573	1678	1575	1328	974	542	240	148		
	RadStd		21	60	76	161	188	131	220	107	110	62	26	11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.83	N/A	-270	N/A	N/A

Nomenclature: See separate page